

PROJECT REPORT

Nmap (Network Mapper) for Cyber Security

Network Discovery, Security Auditing & Vulnerability Assessment

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Abstract

Network security is a fundamental aspect of modern information technology, as organizations increasingly rely on interconnected systems for communication, data storage, and service delivery. Identifying active hosts, open ports, running services, and potential security weaknesses is essential for protecting network infrastructure against cyber threats. This project focuses on Nmap (Network Mapper), a powerful open-source network scanning and security auditing tool widely used by cybersecurity professionals for network discovery, vulnerability assessment, and security analysis.

The primary objective of this project is to study the features, working principles, installation process, and practical applications of Nmap in a controlled and authorized environment. The project demonstrates various scanning techniques, including host discovery, port scanning, service and version detection, operating system fingerprinting, aggressive scanning, and vulnerability assessment using the Nmap Scripting Engine (NSE). Each scanning technique is performed using appropriate Nmap commands, and the results are analysed to understand the security posture of the target system.

A virtual laboratory environment was established using authorized test machines to ensure that all scanning activities were conducted safely and ethically. The observations obtained from each scan provide valuable information about network services, open ports, software versions, operating systems, and potential vulnerabilities that could be exploited if left unaddressed. Based on these findings, appropriate security recommendations are discussed to improve the overall security posture of the target environment.

This project highlights the significance of Nmap as an efficient and versatile tool for network reconnaissance, security auditing, and vulnerability assessment. It demonstrates how regular network scanning can assist organizations in identifying security risks, maintaining accurate asset inventories, verifying firewall configurations, and supporting proactive cybersecurity practices. The knowledge and practical experience gained through this project provide a strong foundation for understanding network security assessment techniques and their role in protecting modern information systems.

Nmap (Network Mapper) for Cyber Security

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Chapter 1: Introduction

Network security has become one of the most critical concerns for organizations of all sizes, as digital infrastructure continues to expand and cyber threats grow increasingly sophisticated. A fundamental requirement of any security program is visibility — an organization cannot protect assets it does not know exist. This project focuses on Nmap (Network Mapper), one of the most widely used open-source tools for network discovery, port scanning, service detection, and vulnerability assessment.

Nmap was designed to allow administrators and security professionals to scan large networks quickly, while also being flexible enough for detailed scans of single hosts. It works by sending specially crafted packets to target systems and analysing the responses to determine which hosts are active, what ports are open, what services are running, and what operating systems are in use.

This project explores Nmap from both a conceptual and practical standpoint — covering its role in cyber security, how it is installed and configured, the methodology behind different scan types, and a hands-on demonstration of its core capabilities against a test environment.

Chapter 2: Problem Statement

Organizations today face significant challenges in maintaining a secure network environment. Some of the key problems that motivate this project include:

- Lack of visibility into network assets, active devices, and running services, which makes it difficult to assess the true attack surface of an organization.
- Inability to detect unauthorized or rogue devices connecting to the network, increasing the risk of unauthorized access and data breaches.
- Unknown or misconfigured services and open ports that may expose the network to exploitation.
- Difficulty in meeting compliance and regulatory requirements (such as GDPR and PCI DSS) that mandate regular network discovery and vulnerability assessment.
- Absence of a low-cost, flexible tool for routine security auditing, especially for smaller teams with limited budgets.

These challenges highlight the need for a reliable, well-understood tool that can be used to continuously map and assess network infrastructure — which is precisely the gap that Nmap is designed to fill.

Chapter 3: Objectives

The objectives of this project are as follows:

- To understand the architecture, features, and working principles of Nmap as a network scanning tool.
- To install and configure Nmap in a controlled lab/test environment.
- To study the methodology behind different types of scans, including host discovery, port scanning, service/version detection, and OS fingerprinting.
- To perform practical scans on a test network and document the commands, outputs, and observations.
- To analyse how Nmap contributes to identifying vulnerabilities and strengthening an organization's security posture.
- To present findings in a structured, professional report format suitable for academic evaluation.

Chapter 4: Nmap Overview

4.1 What is Nmap?

Nmap, short for "Network Mapper," is a free and open-source utility used for network discovery and security auditing. It is widely used by system administrators and security professionals for tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime.

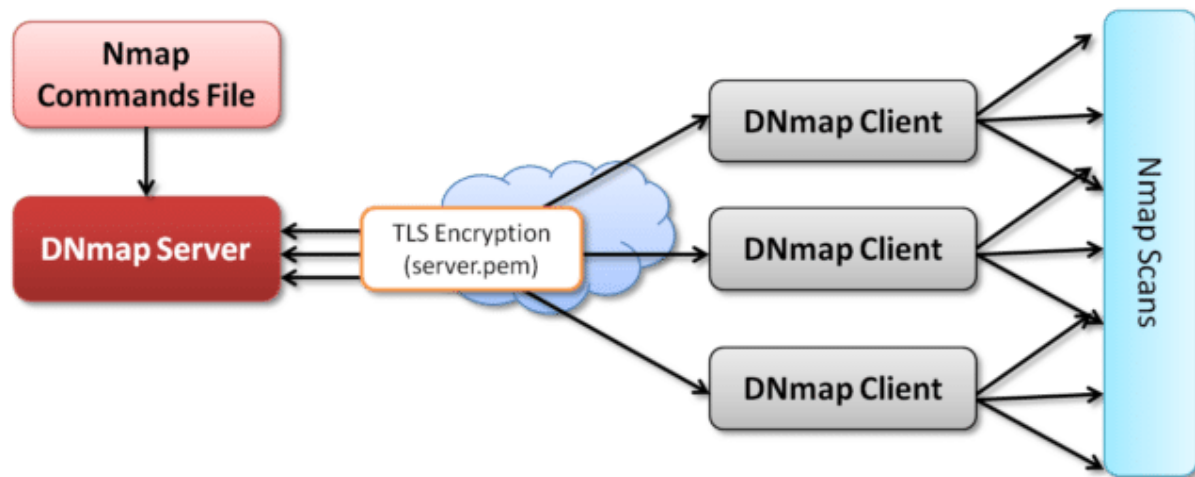


Figure 4.1 Nmap Architecture and Working Process

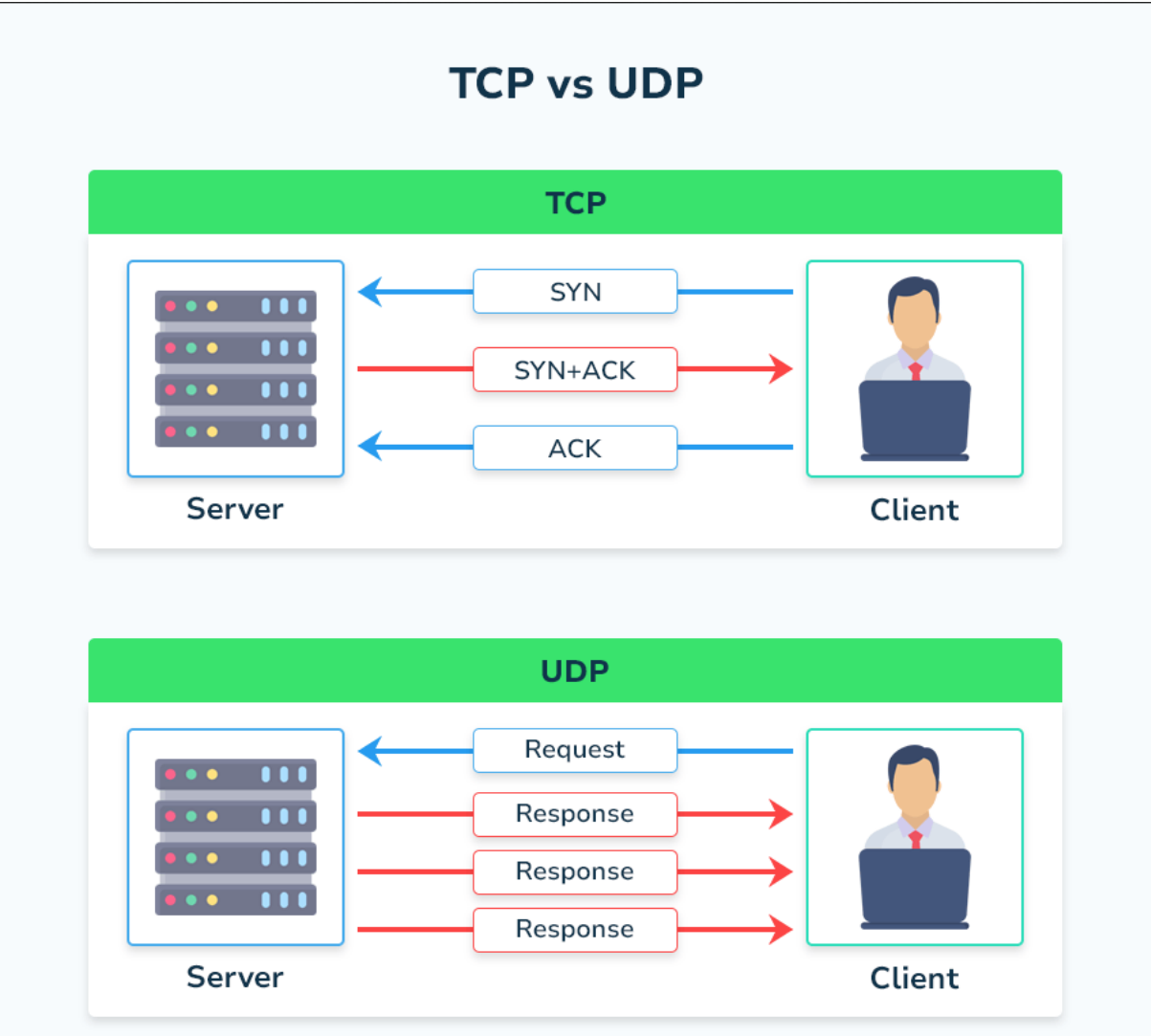
4.2 Key Features

- **Network Discovery:** Identifies active hosts, IP addresses, and devices on a network.
- **Port Scanning:** Detects open, closed, and filtered ports on target systems.
- **Service and Version Detection:** Identifies the services running on open ports and their version numbers.
- **Operating System Detection:** Uses TCP/IP stack fingerprinting to guess the target's operating system.
- **Scripting Engine (NSE):** Allows automation of advanced discovery and vulnerability-detection tasks using Lua scripts.
- **Output and Reporting:** Supports multiple output formats (normal, XML, grepable) for integration with other tools.

4.3 Relevance to Cyber Security

Nmap plays a central role in the reconnaissance and assessment phases of security work. Security teams use it to build an accurate inventory of network assets,

identify potentially vulnerable services before attackers do, and verify that firewall and access-control configurations behave as intended.



Chapter 5: Installation

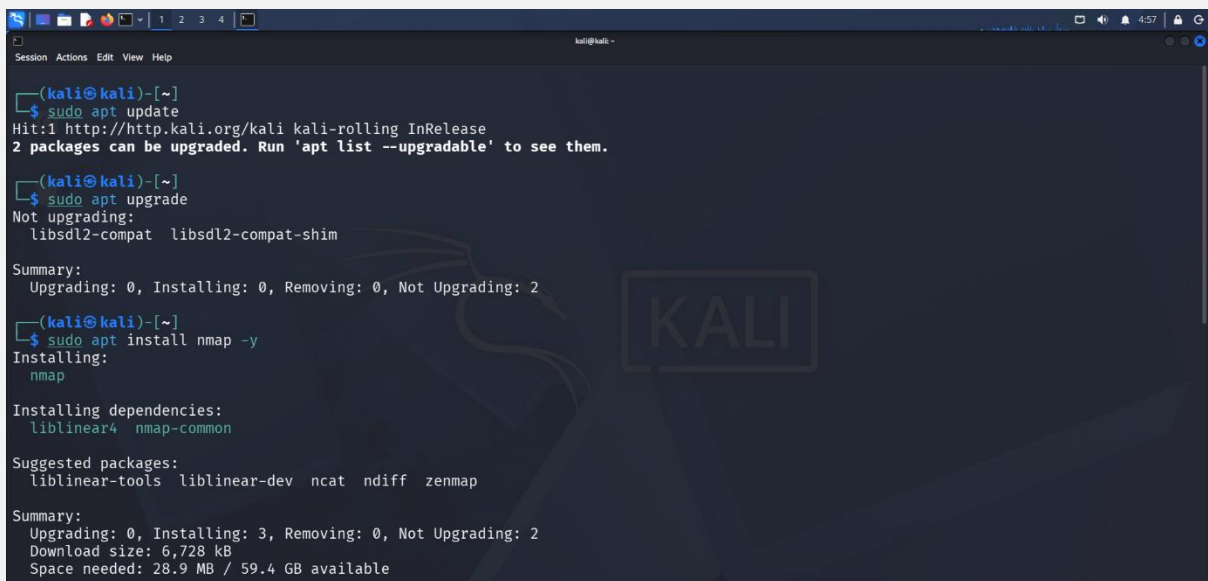
Nmap is available for Windows, Linux, and macOS. The installation steps used for this project are outlined below.

5.1 Installation on Linux

```
$ sudo apt update
```

```
$ sudo apt install nmap -y
```

```
$ nmap --version
```



```
(kali@kali)-[~]
$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)-[~]
$ sudo apt upgrade
Not upgrading:
libstdl2-compat libstdl2-compat-shim

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2

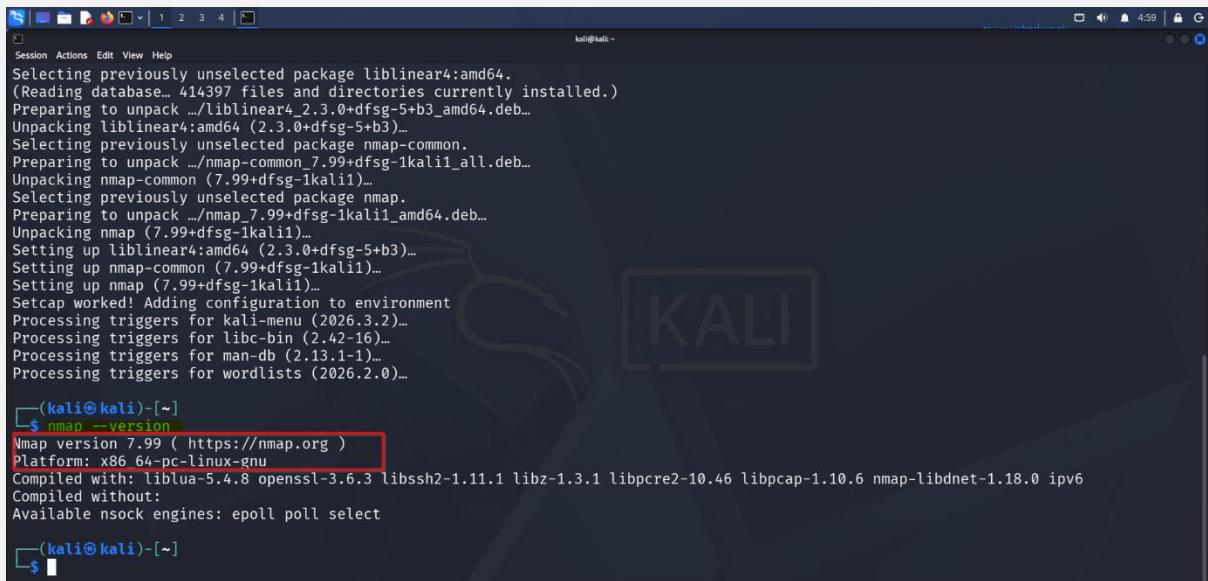
(kali@kali)-[~]
$ sudo apt install nmap -y
Installing:
nmap

Installing dependencies:
liblinear4 nmap-common

Suggested packages:
liblinear-tools liblinear-dev ncat ndiff zenmap

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 2
Download size: 6,728 kB
Space needed: 28.9 MB / 59.4 GB available
```

Figure 5.1 Nmap Installation on Linux Using APT Package Manager



```
Session Actions Edit View Help
Selecting previously unselected package liblinear4:amd64.
(Reading database... 414397 files and directories currently installed.)
Preparing to unpack .../liblinear4_2.3.0+dfsg-5+b3_amd64.deb...
Unpacking liblinear4:amd64 (2.3.0+dfsg-5+b3)...
Selecting previously unselected package nmap-common.
Preparing to unpack .../nmap-common_7.99+dfsg-1kali1_all.deb...
Unpacking nmap-common (7.99+dfsg-1kali1)...
Selecting previously unselected package nmap.
Preparing to unpack .../nmap_7.99+dfsg-1kali1_amd64.deb...
Unpacking nmap (7.99+dfsg-1kali1)...
Setting up liblinear4:amd64 (2.3.0+dfsg-5+b3)...
Setting up nmap-common (7.99+dfsg-1kali1)...
Setting up nmap (7.99+dfsg-1kali1)...
Setcap worked! Adding configuration to environment
Processing triggers for kali-menu (2026.3.2)...
Processing triggers for libc-bin (2.42-16)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for wordlists (2026.2.0)...

(kali@kali)-[~]
$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.3 libssh2-1.11.1 libz-1.3.1 libpcrc2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
$
```

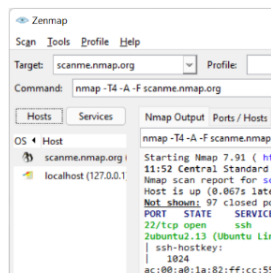
Figure 5.2 Verification of Nmap Installation Using `nmap --version`

5.2 Installation on Windows

On Windows, Nmap is installed using the official installer (Nmap.org), which also includes the Zenmap graphical interface and the Npcap packet-capture driver required for raw packet scanning.

- Download the Windows installer from the official Nmap website.
- Run the installer and accept the license agreement.
- Ensure the Np-cap driver option is selected during installation, as it is required for most scan types.
- Complete the installation and verify by running 'nmap -v' from the Command Prompt.

Microsoft Windows binaries



Please read the [Windows section](#) of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap's dependencies and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 R2 and newer. We also maintain a [guide for users who must run Nmap on earlier Windows releases](#).

Note: The version of Npcap included in our installers may not always be the latest version. If you experience problems or just want the latest and greatest version, download and install [the latest Npcap release](#).

Latest stable release self-installer: [nmap-7.99-setup.exe](#)

Latest Npcap release self-installer: [npcap-1.88.exe](#)

We have written [post-install usage instructions](#). Please [notify us](#) if you encounter any problems or have suggestions for the installer.

Linux RPM Source and Binaries

Many popular Linux distributions (Redhat, Mandrake, Suse, etc) use the [RPM](#) package management system for quick and easy binary package installation. We have written a detailed [guide to installing our RPM packages](#), though these simple commands usually do the trick:

```
rpm -vhU https://nmap.org/dist/nmap-7.99-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/ncat-7.99-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/nping-7.99-1.x86_64.rpm
```

You can also download and install the RPMs yourself:

Latest stable release:

x86-64 (64-bit Linux) Nmap RPM: [nmap-7.99-1.x86_64.rpm](#)

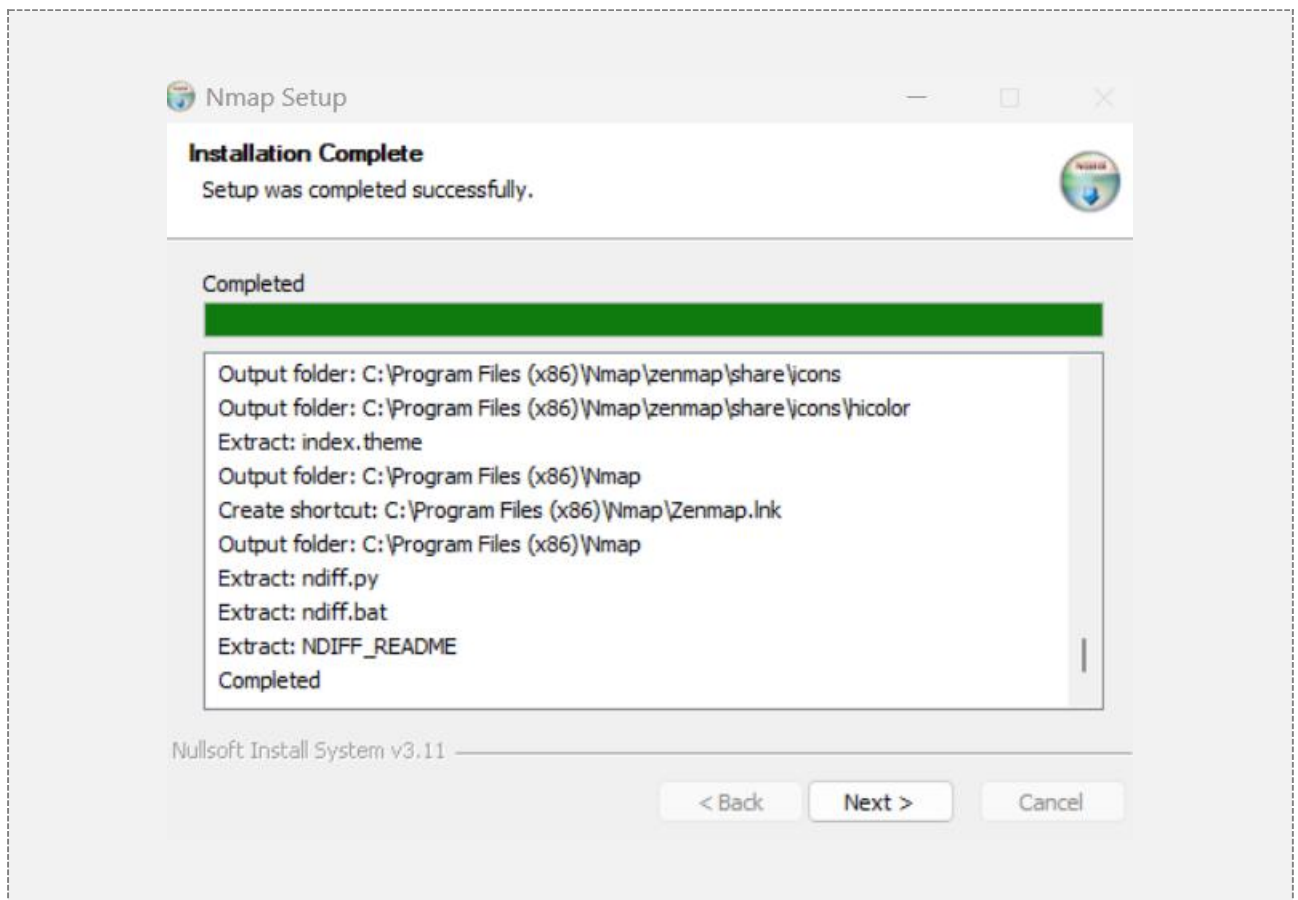
x86-64 (64-bit Linux) Ncat RPM: [ncat-7.99-1.x86_64.rpm](#)

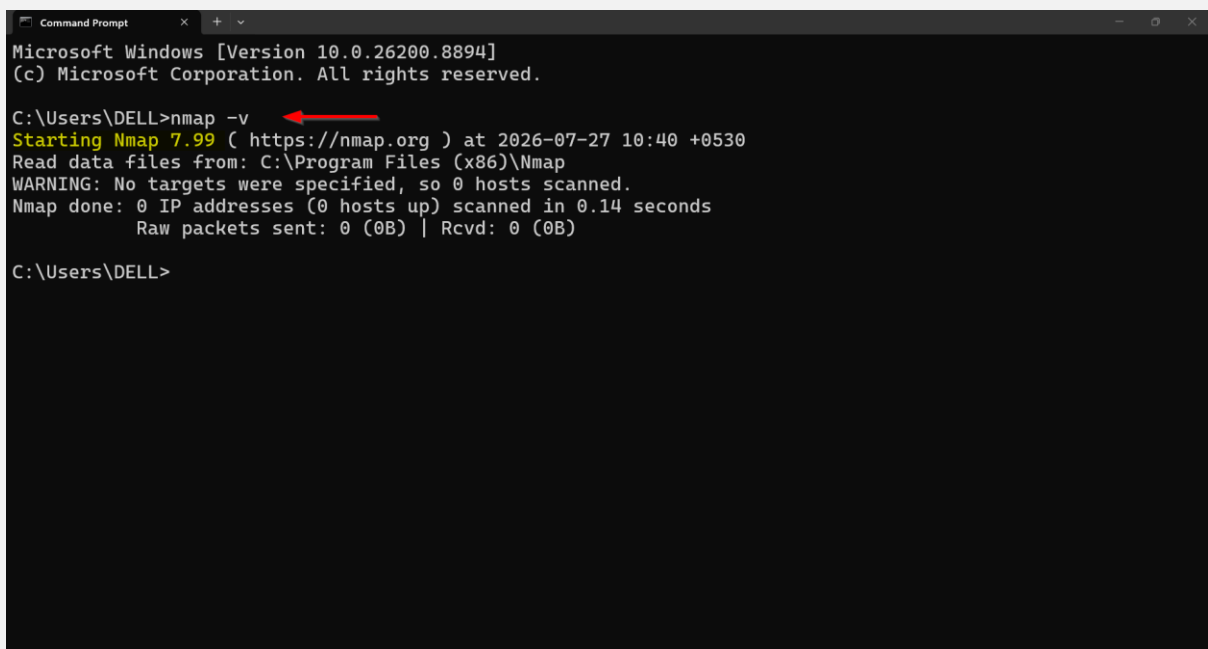
x86-64 (64-bit Linux) Nping RPM: [nping-7.99-1.x86_64.rpm](#)

Optional [Zenmap GUI](#) (all platforms, Python wheel): [zenmap-7.99-py3-none-any.whl](#)

Source RPM (includes Nmap, Zenmap, Ncat, and Nping): [nmap-7.99-1.src.rpm](#)

Figure 5.3 Nmap Installation on Windows.





```
Command Prompt
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 10:40 +0530
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.14 seconds
Raw packets sent: 0 (0B) | Rcvd: 0 (0B)

C:\Users\DELL>
```

Chapter 6: Methodology

The project followed a structured methodology to ensure the scanning activity was conducted safely, ethically, and within a controlled environment.

6.1 Test Environment

All scans in this project were performed only against systems the team owns or is explicitly authorized to test (e.g., a local virtual machine lab or a designated test target). Scanning systems without authorization is illegal and was strictly avoided.

6.2 Scanning Approach

- Step 1 — Host Discovery: Identify which hosts on the target network are active/reachable.
- Step 2 — Port Scanning: Scan discovered hosts to identify open, closed, and filtered ports.
- Step 3 — Service & Version Detection: Determine which services and software versions are running on open ports.
- Step 4 — OS Detection: Attempt to identify the operating system of each target host.
- Step 5 — Vulnerability Scripting: Use the Nmap Scripting Engine (NSE) to check for known vulnerabilities.
- Step 6 — Documentation: Record each command used, capture the output, and analyse the results.

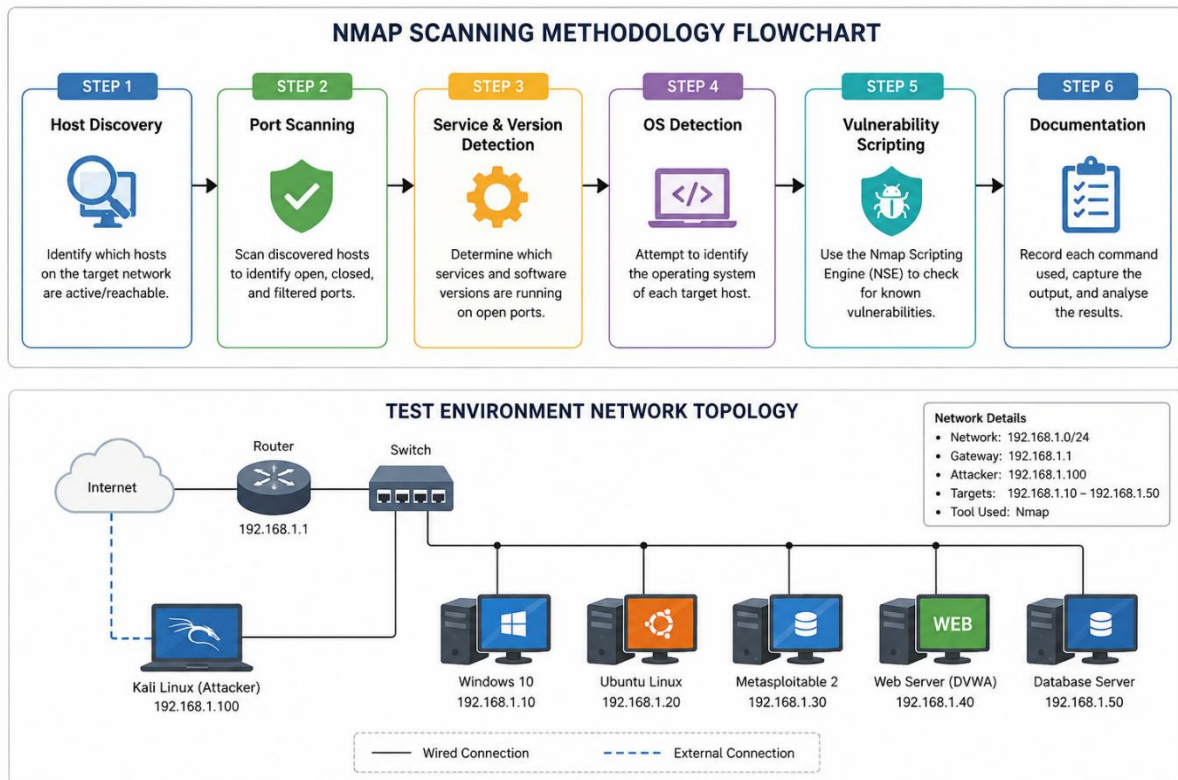


Figure 6.2 Nmap scanning methodology and network topology

6.3 Tools Used

- Nmap (command-line) for scanning.
- A virtual lab environment (e.g., VirtualBox/VMware) with isolated test machines.
- A text editor/terminal for recording commands and outputs.

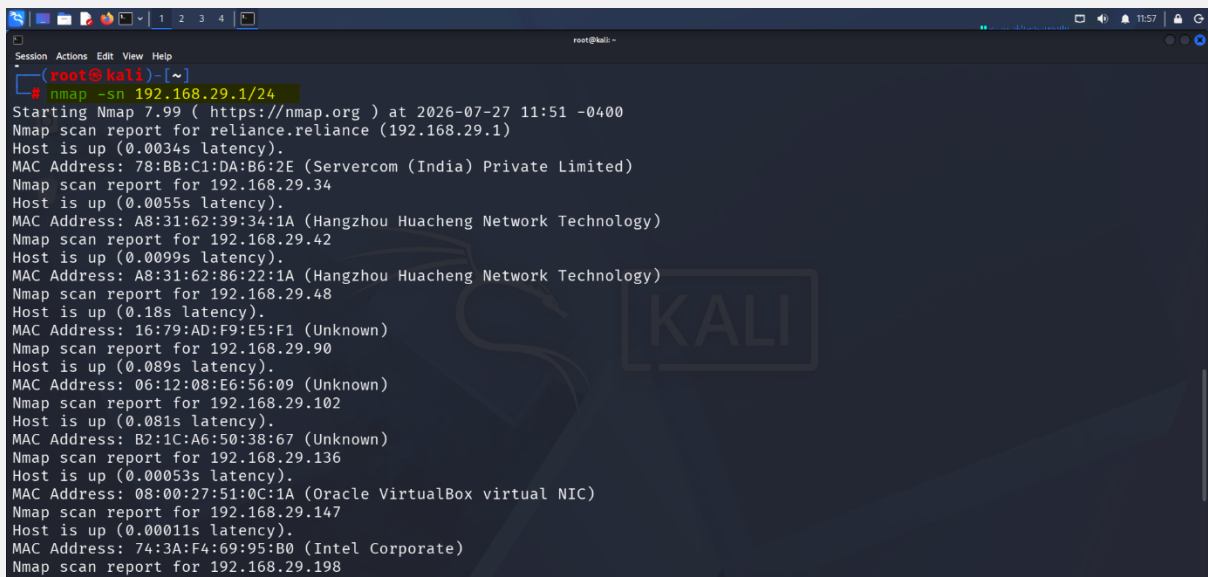
Chapter 7: Practical Demonstration

This chapter presents the hands-on scans performed as part of this project. For each scan type, the command used, the expected purpose, a placeholder for the corresponding screenshot, and observations are provided.

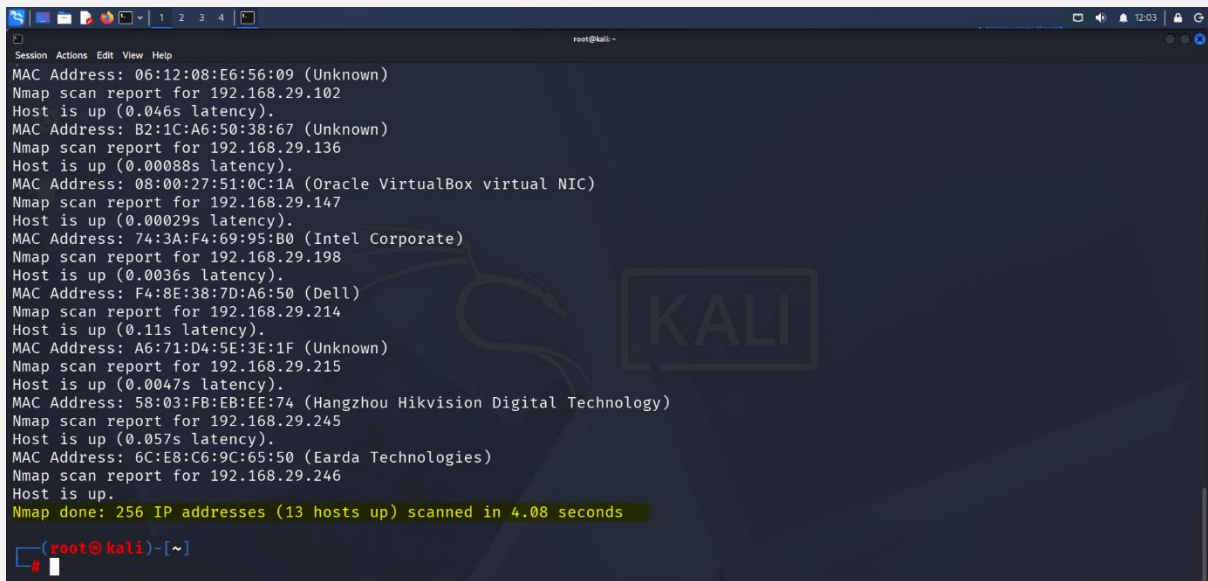
7.1 Host Discovery Scan

Purpose: To identify which hosts on the target subnet are currently active/online.

```
$ nmap -sn 192.168.1.0/24
```



```
root@kali: ~  
$ nmap -sn 192.168.29.1/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 11:51 -0400  
Nmap scan report for reliance.reliance (192.168.29.1)  
Host is up (0.0034s latency).  
MAC Address: 78:BB:C1:DA:B6:2E (Servercom (India) Private Limited)  
Nmap scan report for 192.168.29.34  
Host is up (0.0055s latency).  
MAC Address: A8:31:62:39:34:1A (Hangzhou Huacheng Network Technology)  
Nmap scan report for 192.168.29.42  
Host is up (0.0099s latency).  
MAC Address: A8:31:62:86:22:1A (Hangzhou Huacheng Network Technology)  
Nmap scan report for 192.168.29.48  
Host is up (0.18s latency).  
MAC Address: 16:79:AD:F9:E5:F1 (Unknown)  
Nmap scan report for 192.168.29.90  
Host is up (0.089s latency).  
MAC Address: 06:12:08:E6:56:09 (Unknown)  
Nmap scan report for 192.168.29.102  
Host is up (0.081s latency).  
MAC Address: B2:1C:A6:50:38:67 (Unknown)  
Nmap scan report for 192.168.29.136  
Host is up (0.00053s latency).  
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.29.147  
Host is up (0.00011s latency).  
MAC Address: 74:3A:F4:69:95:B0 (Intel Corporate)  
Nmap scan report for 192.168.29.198
```



```
Session Actions Edit View Help
MAC Address: 06:12:08:E6:56:09 (Unknown)
Nmap scan report for 192.168.29.102
Host is up (0.046s latency).
MAC Address: B2:1C:A6:50:38:67 (Unknown)
Nmap scan report for 192.168.29.136
Host is up (0.00088s latency).
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.29.147
Host is up (0.00029s latency).
MAC Address: 74:3A:F4:69:95:B0 (Intel Corporate)
Nmap scan report for 192.168.29.198
Host is up (0.0036s latency).
MAC Address: F4:8E:38:7D:A6:50 (Dell)
Nmap scan report for 192.168.29.214
Host is up (0.11s latency).
MAC Address: A6:71:D4:5E:3E:1F (Unknown)
Nmap scan report for 192.168.29.215
Host is up (0.0047s latency).
MAC Address: 58:03:FB:EB:EE:74 (Hangzhou Hikvision Digital Technology)
Nmap scan report for 192.168.29.245
Host is up (0.057s latency).
MAC Address: 6C:E8:C6:9C:65:50 (Earda Technologies)
Nmap scan report for 192.168.29.246
Host is up.
Nmap done: 256 IP addresses (13 hosts up) scanned in 4.08 seconds
root@kali:~
```

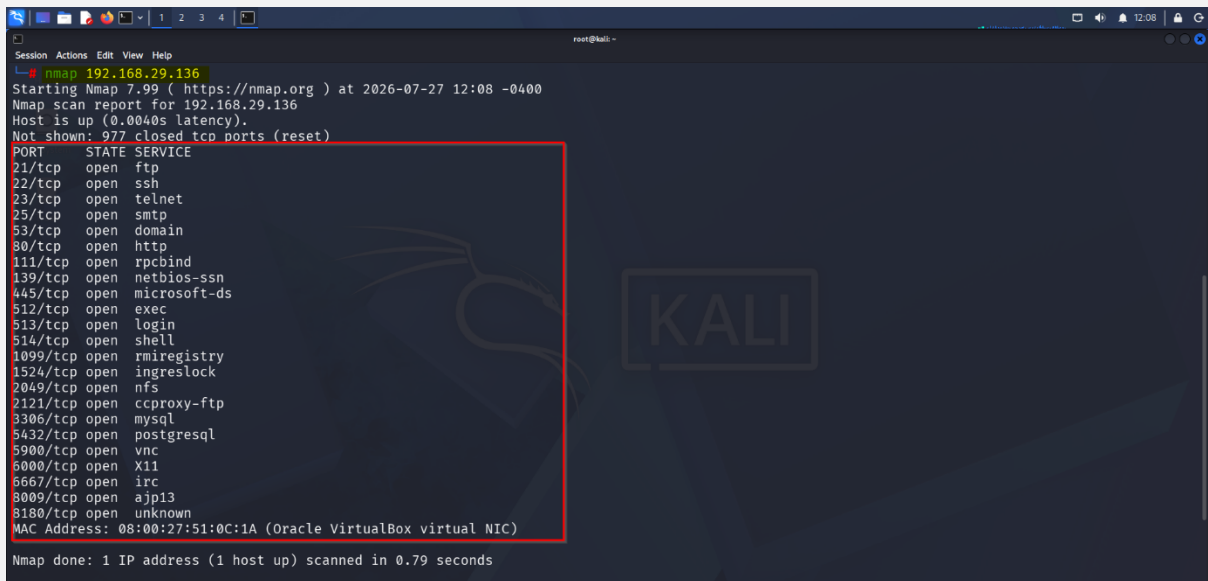
Figure 7.1 Host Discovery Scan Results Using `nmap -sn`

Observation: The “`nmap -sn`” command successfully discovered all active devices on the local network without performing any port scanning. It is commonly used for network inventory, identifying connected hosts, and troubleshooting network connectivity. We have found 13 active devices.

7.2 Basic Port Scan

Purpose: To identify open ports on a specific target host.

```
$ nmap 192.168.1.10
```



```

root@kali: ~
└─$ nmap 192.168.29.136
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 12:08 -0400
Nmap scan report for 192.168.29.136
Host is up (0.0040s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds

```

Figure 7.2 Default TCP Port Scan Results

Observation: The above screenshot displays the results of an “*nmap*” default TCP port scan on the target host 192.168.29.136. The scan successfully identified the host as active and revealed a list of **23 open TCP** ports along with their associated default services, including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), SMB (445), MySQL (3306), PostgreSQL (5432), VNC (5900), X11 (6000), IRC (6667), AJP13 (8009), and others. The remaining **977 TCP ports were reported as closed**. This output provides an overview of the services running on the target system and helps **identify potential entry points** for further security assessment or vulnerability analysis in an authorized environment.

7.3 Service and Version Detection

Purpose: To determine the exact service and version running on each open port.

```
nmap -sV 192.168.1.10
```

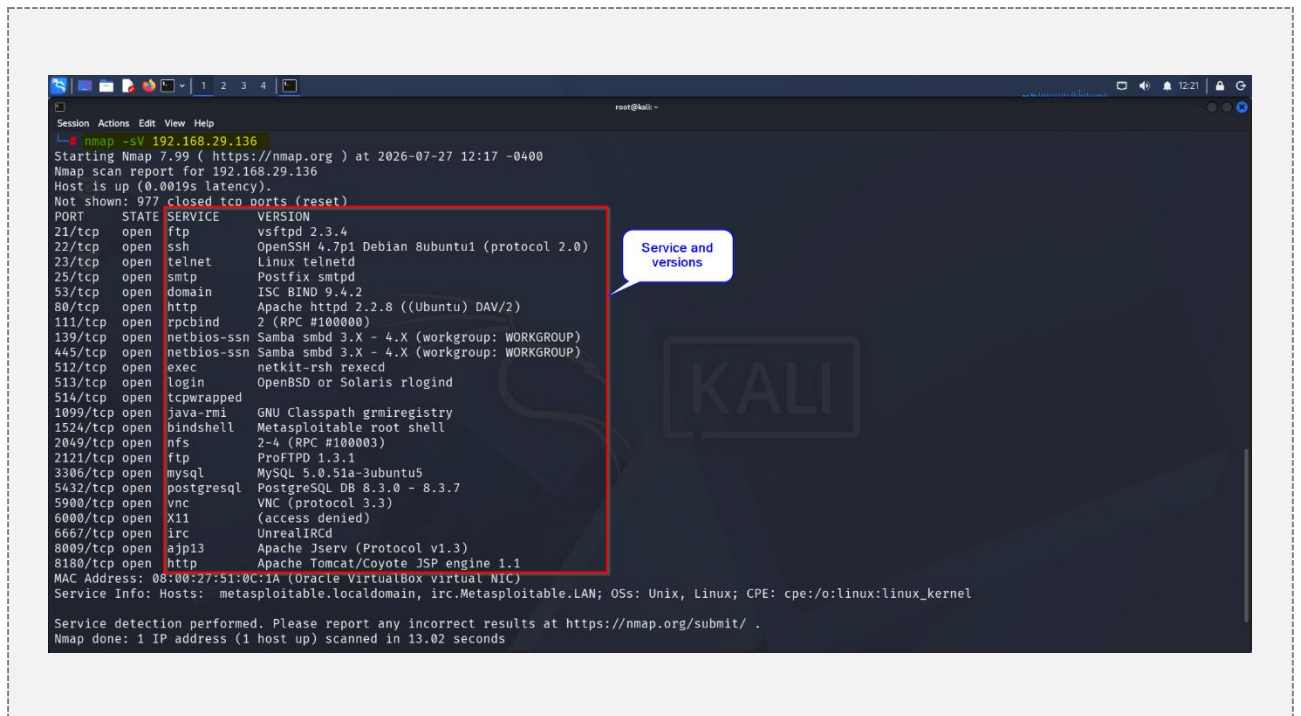


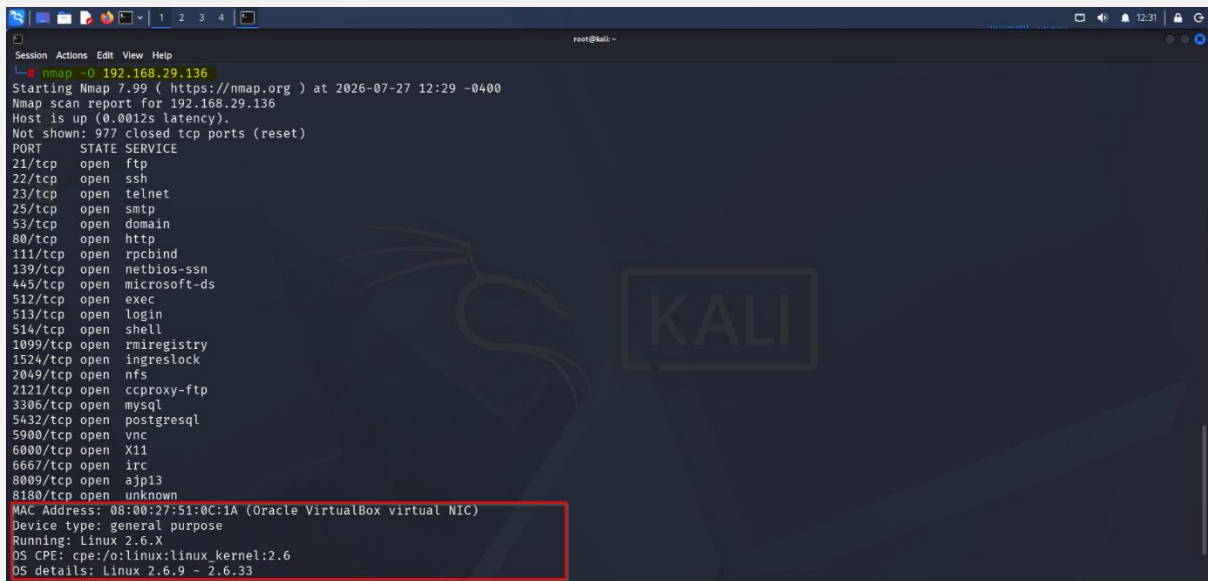
Figure 7.3 Service and Version Detection Results Using `nmap -sV`

Observation: The scan successfully identified the target host as active and detected 23 open TCP ports. In addition to listing the open ports, the ‘-sV’ option identified the service names and their version numbers running on each port. Examples include vsftpd 2.3.4 (FTP), OpenSSH 4.7p1, Apache HTTP Server 2.2.8, ISC BIND 9.4.2 (DNS), Samba 3.x–4.x, MySQL 5.0.51a, PostgreSQL 8.3.x, ProFTPD 1.3.1, Apache Tomcat/Coyote JSP Engine 1.1, and VNC Protocol 3.3. The scan also identified **the target as a Linux/Unix-based Metasploitable system running on an Oracle VirtualBox Virtual NIC**. This information provides detailed insight into the software versions running on the target, which is useful for vulnerability assessment and security analysis.

7.4 Operating System Detection

Purpose: To fingerprint the likely operating system of the target host.

```
nmap -O 192.168.1.10
```



```

Session Actions Edit View Help
nmap -O 192.168.29.136
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 12:29 -0400
Nmap scan report for 192.168.29.136
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccmproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33

```

Figure 7.4 Operating System Detection Results Using nmap -O

Observation: Using TCP/IP fingerprinting, Nmap determined that the target is running a Linux 2.6.x operating system. The OS details indicate a probable kernel version in the range of **Linux 2.6.9 – 2.6.33**, along with the corresponding CPE (Common Platform Enumeration) entry `cpe:/o:linux:linux_kernel:2.6`. The scan also identified the device type as a general-purpose computer, detected the Oracle VirtualBox Virtual NIC as the network interface, and estimated the network distance to be 1 hop. These results provide valuable information about the target system's operating system and network characteristics, supporting further security assessment and vulnerability analysis.

7.5 Aggressive Scan

Purpose: To combine OS detection, version detection, script scanning, and traceroute in a single comprehensive scan.

```
nmap -A 192.168.1.10
```

```

Session Actions Edit View Help
root@kali: ~
nmap -A 192.168.29.136
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 12:35 -0400
Nmap scan report for 192.168.29.136
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.29.246
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC4_128_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5

```

```

Session Actions Edit View Help
root@kali: ~
|_ssl-date: 2026-07-27T16:36:20+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=O
|_COCA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-command: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY,
|_ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind       2 (RPC #100000)
|_rpcinfo:
|_program version  port/proto  service
|_100000 2 111/tcp    rpcbind
|_100000 2 111/udp    rpcbind
|_100003 2,3,4 2049/tcp   nfs
|_100003 2,3,4 2049/udp   nfs
|_100005 1,2,3 42340/tcp  mountd
|_100005 1,2,3 45027/udp  mountd
|_100021 1,3,4 41261/udp  nlockmgr
|_100021 1,3,4 45953/tcp  nlockmgr
|_100024 1 38625/udp  status
|_100024 1 39298/tcp  status
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rshd
513/tcp   open  login         OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)

```

```

Session Actions Edit View Help
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:
|_ <unknown> (unknown)
|_ clock-skew: mean: 1h00m00s, deviation: 2h00m00s, median: 0s
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ smb-security-mode:
|   | account_used: guest
|   | authentication_level: user
|   | challenge_response: supported
|   | message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|   | OS: Unix (Samba 3.0.20-Debian)
|   | Computer name: metasploitable
|   | NetBIOS computer name:
|   | Domain name: localdomain
|   | FQDN: metasploitable.localdomain
|   | System time: 2026-07-27T12:36:13-04:00

TRACEROUTE
HOP RTT ADDRESS
1 1.79 ms 192.168.29.136

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.76 seconds

```

Figure 7.5 Aggressive Scan Results Using nmap -A

Observation: The aggressive scan combined multiple Nmap features, including service version detection, operating system detection, default NSE (Nmap Scripting Engine) scripts, and traceroute. The target host was successfully identified as active with 23 open TCP ports, and detailed information about running services such as vsftpd 2.3.4, OpenSSH 4.7p1, Apache HTTP Server 2.2.8, Samba 3.0.20, MySQL 5.0.51a, PostgreSQL 8.3.x, Apache Tomcat 5.5, and VNC was displayed. Default NSE scripts gathered additional information, including anonymous FTP login availability, SSH host keys, SMTP commands, DNS version, HTTP headers and page titles, MySQL server details, VNC protocol information, SMB configuration, and NetBIOS information. The OS detection identified the target as a Linux 2.6.x system with an estimated kernel version of 2.6.9–2.6.33, running on an Oracle VirtualBox Virtual NIC. The traceroute results indicated the target was 1 network hop away, confirming it was located on the local network. Overall, the aggressive scan provided a comprehensive view of the target's operating system, running services, versions, security-related configurations, and network path, making it highly useful for detailed reconnaissance and vulnerability assessment in an authorized environment.

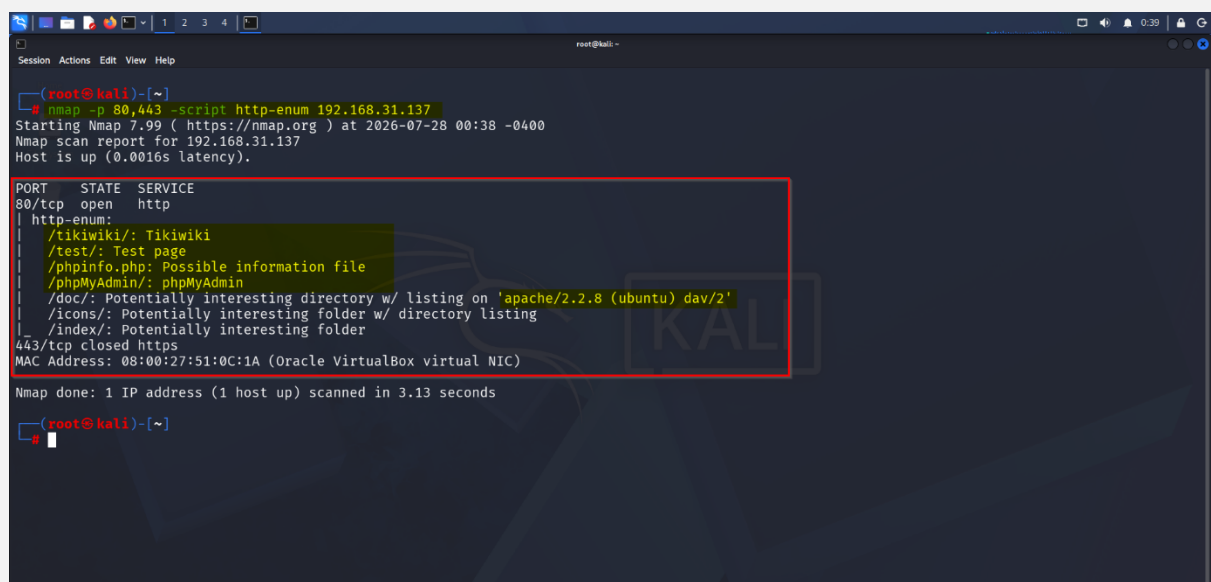
7.6 Advanced Scanning with Nmap Scripting Engine (NSE)

Purpose: To use Nmap's built-in scripting engine to check for known vulnerabilities on the target.

Enumerating Services: NSE scripts for service discovery provide detailed insights.

1. **HTTP Enumeration:** Used to gather detailed information about web servers, their configurations, and hosted applications. It helps identify potential vulnerabilities and areas for further testing.

```
nmap -p 80,443 -script http-enum 192.168.1.10
```



```
(root@kali)~# nmap -p 80,443 -script http-enum 192.168.31.137
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-28 00:38 -0400
Nmap scan report for 192.168.31.137
Host is up (0.0016s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-enum:
| /tikiwiki/: Tikiwiki
| /test/: Test page
| /phpinfo.php: Possible information file
| /phpMyAdmin/: phpMyAdmin
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'
| /icons/: Potentially interesting folder w/ directory listing
| /index/: Potentially interesting folder
443/tcp    closed https
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 3.13 seconds

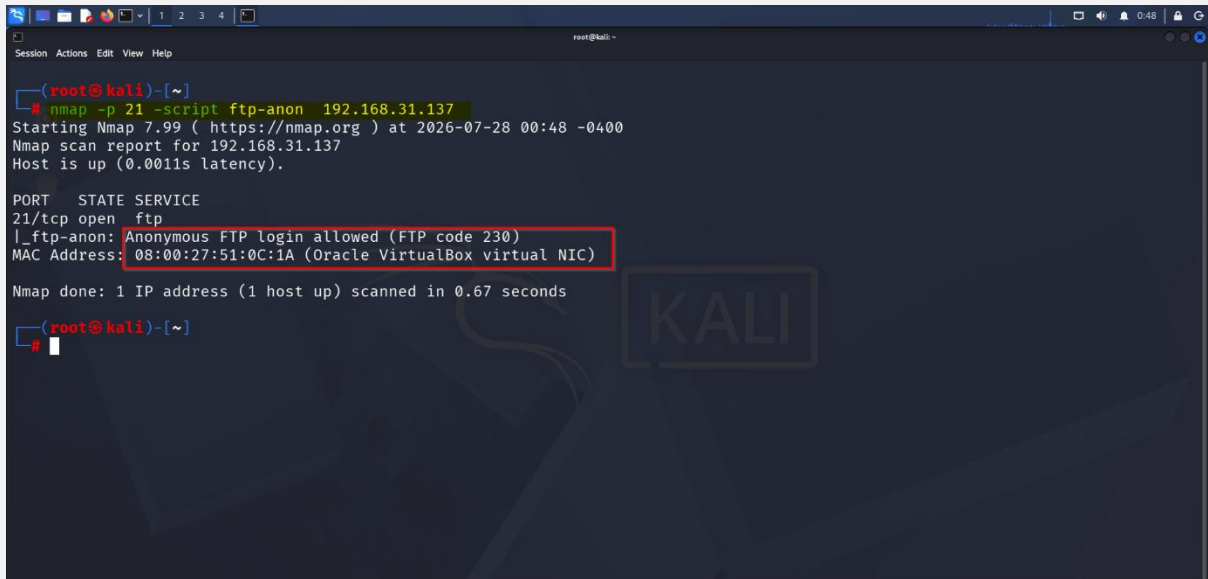
(root@kali)~#
```

Figure 7.6(1) Http Enumeration.

Observation: The HTTP enumeration scan identified several publicly accessible resources, including **TikiWiki**, **phpMyAdmin**, and a **PHP configuration page (phpinfo.php)**. The presence of exposed administrative interfaces, test pages, and enabled directory listings increases the attack surface by providing valuable information about the web server and hosted applications. Additionally, the disclosure of the **Apache 2.2.8 (Ubuntu)** server version may help attackers identify known vulnerabilities associated with outdated software.

- 2. FTP Enumeration:** Used to gather information about an FTP server, such as supported features, anonymous access permissions, and exposed directories. It helps assess potential vulnerabilities in FTP configurations.

```
nmap -p 21 -script ftp-anon,ftp-syn 192.168.1.10
```



```
(root@kali)~  
# nmap -p 21 -script ftp-anon 192.168.31.137  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-28 00:48 -0400  
Nmap scan report for 192.168.31.137  
Host is up (0.0011s latency).  
  
PORT      STATE SERVICE  
21/tcp    open  ftp  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
MAC Address: 08:00:27:51:0C:1A (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds  
  
(root@kali)~  
#
```

Figure 7.6() ftp Enumeration.

Observation: The FTP enumeration scan confirmed that the target system permits **anonymous FTP login**, indicating that authentication is not required to access the FTP service. Such a configuration can expose confidential files, facilitate unauthorized data access, and provide attackers with valuable information for further exploitation. Unless explicitly required for public file distribution, anonymous FTP access should be disabled to reduce the risk of unauthorized access.

Vulnerability Detection:

Vuln: Used to detect vulnerabilities in services, operating systems, and applications running on target devices. It automates the process of vulnerability detection, saving time and ensuring consistency.

```
nmap -script=vuln 192.168.1.10
```

```

root@kali: ~
nmap -script=vuln 192.168.31.137
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-28 00:50 -0400
Nmap scan report for 192.168.31.137
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
ftp-vsftpd-backdoor:
VULNERABLE:
vsFTPD version 2.3.4 backdoor
State: VULNERABLE (Exploitable)
IDs: CVE:CVE-2011-2523 BID:48539
vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
Disclosure date: 2011-07-03
Exploit results:
Shell command: id
Results: uid=0(root) gid=0(root)
References:
http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
https://www.securityfocus.com/bid/48539
https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
sslv2-drown: ERROR: Script execution failed (use -d to debug)
sslv2-poodle:
VULNERABLE:
SSL POODLE information leak
State: VULNERABLE
sslv2-dh-params:
VULNERABLE:
Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
State: VULNERABLE
Transport Layer Security (TLS) services that use anonymous
Diffie-Hellman key exchange only provide protection against passive
eavesdropping, and are vulnerable to active man-in-the-middle attacks
which could completely compromise the confidentiality and integrity
of any data exchanged over the resulting session.
Check results:
ANONYMOUS DH GROUP 1
Cipher Suite: TLS_DH_anon_WITH_RC4_128_MD5
Modulus Type: Safe pTime
Modulus Source: postfix builtin
Modulus Length: 1024
Generator Length: 8
Public Key Length: 1024
References:
https://www.ietf.org/rfc/rfc2246.txt
Transport Layer Security (TLS) Protocol DHE_EXPORT Ciphers Downgrade MitM (Logjam)
State: VULNERABLE
IDs: CVE:CVE-2015-4000 BID:74733
The Transport Layer Security (TLS) protocol contains a flaw that is
triggered when handling Diffie-Hellman key exchanges defined with
the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
to downgrade the security of a TLS session to 512-bit export-grade
cryptography, which is significantly weaker, allowing the attacker
to more easily break the encryption and monitor or tamper with
the encrypted stream.
Disclosure date: 2015-5-19
Check results:

```

```

Session Actions Edit View Help
https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2015-4000
Diffie-Hellman Key Exchange Insufficient Group Strength
State: VULNERABLE
Transport Layer Security (TLS) services that use Diffie-Hellman groups
of insufficient strength, especially those using one of a few commonly
shared groups, may be susceptible to passive eavesdropping attacks.
Check results:
WEAK DH GROUP 1
Cipher Suite: TLS_DHE_RSA_WITH_DES_CBC_SHA
Modulus Type: Safe prime
Modulus Source: postfix builtin
Modulus Length: 1024
Generator Length: 8
Public Key Length: 1024
References:
https://weakdh.org
smtp-vuln-cve2010-4344:
The SMTP server is not Exim: NOT VULNERABLE
53/tcp open domain
80/tcp open http
http-csrf:
Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.31.137
Found the following possible CSRF vulnerabilities:

Path: http://192.168.31.137:80/dvwa/
Form id:
Form action: login.php

Path: http://192.168.31.137:80/dvwa/login.php
Form id:
Form action: login.php

Session Actions Edit View Help
SSL-CCS-Injection:
VULNERABLE:
SSL/TLS MITM vulnerability (CCS Injection)
State: VULNERABLE
Risk factor: High
OpenSSL before 0.9.8za, 1.0.0 before 1.0.0m, and 1.0.1 before 1.0.1h
does not properly restrict processing of ChangeCipherSpec messages,
which allows man-in-the-middle attackers to trigger use of a zero
length master key in certain OpenSSL-to-OpenSSL communications, and
consequently hijack sessions or obtain sensitive information, via
a crafted TLS handshake, aka the "CCS Injection" vulnerability.
References:
http://www.openssl.org/news/secadv_20140605.txt
http://www.cvedetails.com/cve/2014-0224
https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-0224
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
| irc-unrealircd-backdoor: Looks like trojaned version of unrealircd. See http://seclists.org/fulldisclosure/2010/Jun/277
8009/tcp open ajp13
8180/tcp open unknown
http-cookie-flags:
/admin/:
JSESSIONID:
httponly flag not set
/admin/index.html:
JSESSIONID:
httponly flag not set
/admin/login.html:
JSESSIONID:
httponly flag not set

```

Figure 7.6(3) Vulnerability Assessment Using Nmap Scripting Engine (-script=vuln)

Observation: The vulnerability assessment performed using the Nmap NSE vuln script identified multiple critical and high-risk vulnerabilities on the target machine (Metasploitable 2). Several outdated and intentionally vulnerable services were detected, including vulnerable FTP, SSL/TLS misconfigurations, weak cryptographic protocols, and insecure web application settings.

7.7 Nmap NSE Vulnerability Assessment Results

Table: Vulnerabilities Detected During NSE Vulnerability Scan

S.No	Vulnerability Detected	Port/Service	Severity	Potential Impact
1	vsFTPD 2.3.4 Backdoor (CVE-2011-2523)	FTP (21)	Critical	Allows attackers to gain unauthorized root access through the backdoored FTP service, leading to complete system compromise.
2	SSL POODLE Attack	SSL/TLS	High	Enables attackers to decrypt encrypted communications by exploiting SSLv3 protocol weaknesses.
3	Anonymous Diffie-Hellman Key Exchange (MITM Vulnerability)	SSL/TLS	High	Allows Man-in-the-Middle attacks due to anonymous authentication during TLS handshake.
4	Logjam Attack (CVE-2015-4000)	SSL/TLS	High	Weak export-grade Diffie-Hellman parameters allow attackers to downgrade encryption and intercept secure communications.
5	Weak Diffie-Hellman Group Strength	SSL/TLS	Medium–High	Uses 1024-bit DH parameters which are considered weak and susceptible to cryptographic attacks.

6	SSL CCS Injection (CVE-2014-0224)	SSL/TLS	High	Permits session hijacking and Man-in-the-Middle attacks by exploiting improper ChangeCipherSpec processing.
7	UnrealIRCd Backdoor Detection	IRC (6667)	Critical	Indicates the IRC server contains a trojaned version capable of executing arbitrary commands remotely.
8	HTTP CSRF Vulnerability	HTTP (80)	Medium	The web application (DVWA) may allow Cross-Site Request Forgery attacks, enabling unauthorized actions on behalf of authenticated users.
9	Missing HttpOnly Cookie Flag	HTTP	Medium	Session cookies can potentially be accessed through client-side scripts, increasing the risk of session theft via XSS.

7.8 Summary of Findings

The scan reveals that the target system contains **multiple intentionally vulnerable services**, making it highly susceptible to exploitation. The most critical findings include the **vsFTPd 2.3.4 backdoor** and the **UnrealIRCd backdoor**, both of which can provide attackers with remote command execution and full system compromise. Additionally, several outdated SSL/TLS configurations (POODLE, Logjam, CCS Injection, Weak DH Parameters, and Anonymous DH) expose encrypted communications to interception and cryptographic attacks. The presence of **CSRF vulnerabilities** and **missing HttpOnly cookie flags** further indicates weaknesses in the web application's security posture.

Overall Risk Rating: Critical. Immediate remediation would be required in a production environment by updating vulnerable software, disabling insecure protocols, strengthening cryptographic configurations, and implementing secure web application practices.

Chapter 8: Risk Analysis

8.1 Overview

The Nmap scans conducted during this project identified several active services and open ports on the target system. While the purpose of the scans was to assess the security posture of an authorized test environment, the results demonstrate how exposed network services can present potential security risks if they are not properly configured, monitored, or updated.

8.2 Risk Identified

Risk Identified	Potential Impact	Severity	Recommended Mitigation
Open FTP Service (Port 21)	Unauthorized file access or anonymous login	High	Disable anonymous access and use SFTP.
Open Telnet Service (Port 23)	Credentials transmitted in plain text	Critical	Replace Telnet with SSH.
Outdated Service Versions	Known vulnerabilities	High	Regularly update and patch services.
Multiple Open Services	Increased attack surface	Medium	Disable unnecessary services and restrict access.
Potential Vulnerabilities Detected by NSE	Known security issues	High	Apply patches and verify by rescanning.

8.3 Risk Assessment

The practical scans demonstrated that systems exposing multiple services require continuous monitoring and periodic security assessments. Regular network discovery and vulnerability scanning help identify risks before they are exploited.

8.4 Security Recommendations

- Perform regular network scanning.
- Disable unused ports and unnecessary services.
- Replace insecure protocols such as Telnet with SSH.
- Keep systems and applications updated.
- Configure firewalls appropriately.
- Conduct periodic vulnerability assessments.
- Maintain documentation of scan results and remediation.

Chapter 9: Conclusion

This project provided a comprehensive understanding of Nmap (Network Mapper) and its role in network discovery, security auditing, and vulnerability assessment. Practical implementation demonstrated host discovery, port scanning, service and version detection, operating system fingerprinting, aggressive scanning, and vulnerability assessment using the Nmap Scripting Engine (NSE).

The scans highlighted the importance of identifying exposed services, outdated software, and insecure configurations. Appropriate remediation measures such as patch management, service hardening, firewall configuration, and continuous monitoring can significantly improve an organization's security posture.

Overall, Nmap is an essential tool for cybersecurity professionals, network administrators, penetration testers, and security auditors. When used responsibly and with proper authorization, it supports proactive security assessments and strengthens network defenses. This project has enhanced our practical knowledge of cybersecurity and provides a strong foundation for future work in penetration testing and vulnerability management.